

Question: How would you find the average price of all the products sold in the past month?
Assume 3x3 matrix with each row representing the sales for a different product

Aim

To calculate the **average price of all products sold** using a NumPy array.

Algorithm

1. Import the NumPy library.
2. Create a **3 × 3 NumPy array**, where each row represents a different product.
3. Store the sales/price values in the array.
4. Use `np.mean()` to calculate the average of all values in the array.
5. Display the calculated average price.

Python Code

```
import numpy as np

# 3x3 matrix representing sales data for 3 products
sales_data = np.array([
    [100, 120, 150],
    [200, 180, 220],
    [80, 90, 110]
])

# Calculate average price of all products sold
average_price = np.mean(sales_data)

print("Sales Data:")
print(sales_data)

print("\nAverage Price of All Products:", average_price)
```

Input

100 120 150

200 180 220

80 90 110

Output

Sales Data:

```
[[100 120 150]
```

```
[200 180 220]
```

```
[ 80  90 110]]
```

Average Price of All Products: 141.11111111111111

Result

Thus, the **average price of all products sold is 141.11**.

```
= RESTART: C:/Users/Madaraboina Gayathri/OneDrive/Desktop/DSA0402-FODS/EX-2.py =  
Sales Data:  
[[100 120 150]  
 [200 180 220]  
 [ 80  90 110]]  
  
Average Price of All Products: 138.88888888888889
```